RELIABILITY-BASED DESIGN OPTIMIZATION OF OFFSHORE WIND TURBINE SUPPORT STRUCTURES WITH ENHANCED FATIGUE ASSESSMENT

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***Abstract*—This paper presents a reliability-based design optimization (RBDO) framework for offshore wind turbine (OWT) support structures using Radial Basis Function (RBF) neural networks as surrogate models. The methodology addresses the computational challenges of traditional RBDO by replacing expensive finite element analyses with efficient RBF approximations. A 5 MW monopile OWT support structure located in Kalpakkam, India, is optimized considering six limit states: maximum von Mises stress, buckling stability, tower top displacement, mudline rotation, natural frequency constraints, and fatigue damage accumulation. The design variables include diameters and thicknesses of tower, transition piece, and monopile sections. Using Latin Hypercube Sampling, 360 finite element simulations generate training data for RBF models. Particle Swarm Optimization performs both deterministic design optimization (DDO) and RBDO. Results demonstrate that RBDO achieves a system reliability index of 3.18 compared to 2.76 for DDO, with only 5.6% volume increase. Sensitivity analysis reveals that soil uncertainty significantly influences optimal designs, with higher coefficient of variation requiring more conservative dimensions. The proposed framework provides an efficient and practical approach for reliability-based structural design of offshore wind installations.**

***Index Terms*—Offshore wind turbine, Reliability-based design optimization, Radial basis function, Surrogate modeling, Fatigue analysis**

I. Introduction

A. Background and Motivation

The global transition toward renewable energy has positioned offshore wind power as a critical component of sustainable energy infrastructure. As of 2024, offshore wind installations have exceeded 60 GW globally, with projections indicating substantial growth in the coming decades. The support structures of offshore wind turbines constitute approximately 25-30% of total project costs, making their optimization economically significant. However, these structures operate in harsh marine environments characterized by complex loading combinations from wind, waves, currents, and seismic activities, necessitating robust design methodologies that account for inherent uncertainties.

Traditional deterministic design optimization approaches apply conservative safety factors to accommodate uncertainties, often resulting in over-designed structures. While such approaches ensure safety, they lead to material inefficiency and increased costs. Reliability-based design optimization offers a rational framework that explicitly considers uncertainties in loads, material properties, and geometric parameters while maintaining specified safety levels through probabilistic constraints. However, conventional RBDO implementations require numerous structural analyses within nested optimization loops, creating prohibitive computational burdens for complex finite element models of OWT support structures.

B. Literature Review

Several researchers have addressed the optimization of OWT support structures using various approaches. Zwick and Muskulus performed reliability-based design of monopile foundations using simplified analytical models, demonstrating the potential for material savings. Sørensen and Toft developed probabilistic design frameworks for OWT components considering multiple failure modes. However, these studies often relied on simplified structural models or limited uncertainty characterization.

The integration of surrogate modeling techniques with RBDO has emerged as an effective strategy to reduce computational costs. Polynomial chaos expansions, Kriging models, and artificial neural networks have been employed to approximate structural responses. Among these, Radial Basis Function networks offer advantages in terms of accuracy for highly nonlinear responses and efficiency in high-dimensional design spaces. Recent applications of RBF networks in structural optimization have demonstrated their capability to capture complex response surfaces with relatively modest training datasets.

Fatigue assessment of OWT support structures presents particular challenges due to the stochastic nature of environmental loading and the requirement for long-term damage accumulation estimates. Studies by Velarde et al. highlighted the importance of incorporating fatigue limit states in OWT optimization. Despite these advances, comprehensive RBDO

frameworks that simultaneously address multiple limit states including fatigue damage remain limited in the literature.

This study develops an efficient RBDO methodology for OWT monopile support structures by integrating RBF surrogate models with particle swarm optimization. The specific objectives are:

1) Develop accurate RBF surrogate models for six critical structural response quantities: von Mises stress, buckling load multiplier, displacement, rotation, natural frequency, and fatigue damage.
2) Implement both deterministic and reliability-based optimization using particle swarm optimization to compare design outcomes.
3) Evaluate the influence of soil uncertainty on optimal designs through parametric sensitivity analysis.
4) Demonstrate the practical applicability of the framework using site-specific environmental data from Kalpakkam, India.

The scope encompasses a 5 MW offshore wind turbine with monopile support structure installed in 20 m water depth. Six design variables representing diameters and wall thicknesses of tower, transition piece, and monopile sections are optimized to minimize structural volume while satisfying probabilistic constraints with a target reliability index of 3.71, corresponding to a failure probability of approximately 10^{-4} per year.

The remainder of this paper is organized as follows: Section II describes the structural configuration and environmental loading conditions. Section III presents the RBDO formulation and RBF surrogate modeling approach. Section IV details the optimization methodology using particle swarm optimization. Section V presents comprehensive results including surrogate model validation, optimization outcomes, and reliability assessments. Section VI discusses implications and limitations. Finally, Section VII provides conclusions and recommendations for future work.

Structural Configuration and Loading

C. Offshore Wind Turbine System

The study considers a 5 MW horizontal-axis offshore wind turbine with a rotor diameter of 126 m and a hub height of 90 m above mean sea level. The turbine operates at a rated wind speed of 11.4 m/s, with cut-in and cut-out wind speeds of 3 m/s and 25 m/s, respectively. The rotor–nacelle assembly (RNA) mass is 350,000 kg, and the turbine functions within an operating rotational speed range of 6.9–12.1 rpm.

The support structure consists of three primary components:

1) **Tower:** Extending from 20 m above MSL to hub height (77.6 m length)
2) **Transition piece:** Connecting tower to monopile (10 m length)
3) **Monopile foundation:** Extending from seabed through water column (56 m length, 36 m embedment depth)

Material properties for structural steel are: Young's modulus $E = 210$ GPa, yield strength $\sigma_y = 355$ MPa, Poisson's ratio $\nu = 0.3$, and effective density (accounting for buoyancy) $\rho_{eff} = 8500$ kg/m³.

D. Environmental Conditions

The site is located off the coast of Kalpakkam, Tamil Nadu, India (12.55°N, 80.17°E), in the Bay of Bengal with a water depth of 20 m. Environmental data spanning 20 years (1998-2017) inform the design parameters.

Wind Loading: The wind profile follows a power-law distribution shown using equation (1):

$$u(z) = u_{hub}\left(\frac{z}{z_{hub}}\right)^{\alpha} \tag{1}$$

where $u(z)$ is the mean wind speed at height z, u_{hub} is the reference wind speed at hub height z_{hub}, and α is the wind shear exponent. For offshore conditions, $\alpha = 0.115$. The 50-year return period wind speed at hub height is 50 m/s. Wind pressure on the tower is calculated through equation (2):

$$P_w = 0.5 C_D \rho_{air} u^2 \tag{2}$$

with drag coefficient $C_D = 1.0$ and air density $\rho_{air} = 1.225$ kg/m³.

Wave and Current Loading: Wave characteristics are derived from long-term hindcast data, which represent numerically reconstructed historical wave and current conditions obtained using validated atmospheric and oceanographic models. Hindcast datasets are widely used in offshore engineering studies to estimate extreme and operational sea states at locations where long-term measured data are limited. For the present study, environmental conditions corresponding to a 50-year return period are considered. The significant wave height is $H_s = 6.9$ m and the peak wave period is $T_s = 7.7$ s. The surface current velocity is taken as $v_c = 0.8$ m/s.

Wave kinematics are computed using linear Airy wave theory, with wave number $k = 2\pi/L$ and angular frequency $\omega = 2\pi/T_s$. Hydrodynamic forces acting on the monopile are evaluated using Morison's equation (3):

$$F = F_{inertia} + F_{drag} = \frac{\pi}{4}\rho_w C_M D^2 \dot{u} + \frac{1}{2}\rho_w C_D D u|u| \tag{3}$$

where F is the total hydrodynamic force per unit length acting on the monopile, composed of an inertia force component $F_{inertia}$ and a drag force component F_{drag}. The inertia term represents the force induced by the acceleration of the surrounding water particles, while the drag term accounts for viscous resistance due to the relative velocity between the fluid and the structure. In the equation, ρ_w denotes the seawater density, C_M is the inertia coefficient, C_D is the drag coefficient, D is the monopile diameter, u is the total horizontal water particle velocity, and $\dot{u}$ is the corresponding water particle acceleration.

E. Load Cases and Combinations

The analysis considers ultimate limit state (ULS) and fatigue limit state (FLS) load combinations:

ULS Load Combination: The ultimate limit state considered in this study includes extreme wind conditions with a

wind speed of 50 m/s at hub height, along with a 50-year return period extreme wave characterized by a significant wave height $H_s = 6.9$ m and a peak wave period $T_s = 7.7$ s. The environmental loading also accounts for a current velocity of 0.8 m/s, in addition to the self-weight of the structure and the rotor–nacelle assembly (RNA) mass. Partial safety factors are applied in the ULS evaluation, with a load factor $\gamma_f = 1.35$ and a material factor $\gamma_m = 1.05$.

FLS Assessment: Fatigue damage is estimated using Palmgren-Miner's rule with S-N curves for welded steel ($m = 3.0$). The annual number of loading cycles is approximated as $0.8 \times 365 \times 24 \times 3600/T_s$, accounting for operational availability.

F. Total Loading Summary

The integrated environmental loads at the mudline, located 20 m below mean sea level (MSL), include axial force components of $F_{Ax} = 46,990$ N, $F_{Ay} = 475,000$ N, and $F_{Az} = 72,670$ N. The corresponding moment components at the mudline are $M_{Ax} = -1.129$ MN·m, $M_{Ay} = -0.113$ MN·m, and $M_{Az} = -0.218$ MN·m. The integrated horizontal wave force acting over the submerged portion of the monopile is $F_{Hx} = 285$ kN, producing an overturning moment of $M_{Hy} = 1.85$ MN·m. In addition, the hydrostatic pressure at the mudline is 196.5 kPa, and the gravitational load due to the rotor–nacelle assembly (RNA) is 3.43 MN. These combined loads form the basis for the finite element simulations used to generate the training data for the surrogate models.

II. Methodology

This section outlines the methodology adopted for the reliability-based design optimization (RBDO) of the offshore wind turbine monopile support structure. The approach combines finite element analysis, probabilistic modeling of uncertainties, and surrogate modeling using radial basis functions to reduce computational effort. Particle swarm optimization is employed to obtain optimal designs that minimize structural volume while satisfying prescribed reliability constraints.

A. RBDO Problem Formulation

The reliability-based design optimization problem is formulated as:

$$\begin{aligned} \text{Minimize: } & f(\boldsymbol{d}) = V(\boldsymbol{d}) = \sum V_i(D_i, T_i, L_i) \\ \text{Subject to: } & P[g_j(\boldsymbol{d}, \boldsymbol{X}) \leq 0] \leq P_{f,target}, \quad j = 1, \ldots, n_g \\ & \beta_j(\boldsymbol{d}) \geq \beta_{target}, \quad j = 1, \ldots, n_g \\ & \boldsymbol{d}^L \leq \boldsymbol{d} \leq \boldsymbol{d}^U \end{aligned} \tag{4}$$

where the design vector is defined as $\boldsymbol{d} = [D_1, D_2, D_3, T_1, T_2, T_3]^T$, representing the diameters and thicknesses of the structural components. The objective function is to minimize the total structural volume $V(\boldsymbol{d})$. Reliability constraints are imposed in terms of either a probability of failure $P_f \leq 10^{-4}$ or an equivalent reliability index $\beta \geq 3.71$. The random variable vector is denoted by $\boldsymbol{X}$, which includes uncertainties in loads, material properties, and soil parameters. The limit state functions g_j represent the governing structural constraints.

B. Design Variables and Bounds

Six design variables with specified bounds are presented in Table I.

TABLE I
Design Variables and Bounds

Variable	Description	Lower Bound	Upper Bound
D_1	Tower top diameter	3.0 m	4.5 m
D_2	Tower base/transition diameter	5.0 m	7.0 m
D_3	Monopile diameter	5.0 m	7.0 m
T_1	Tower wall thickness	12 mm	25 mm
T_2	Transition wall thickness	20 mm	35 mm
T_3	Monopile wall thickness	40 mm	70 mm

Additional geometric constraints ensure manufacturability: $D_1 \leq D_2 \leq D_3$, $T_1 \leq T_2 \leq T_3$, and $D_3/T_3 \leq 120$ (slenderness limit).

C. Limit State Functions

Six critical limit states are considered in the reliability-based design optimization of the monopile support structure. The first limit state, g_1, governs the maximum von Mises stress, defined as:

$$g_1 = \sigma_{yield} - \sigma_{vm,max}(\boldsymbol{d}, \boldsymbol{X}) \tag{5}$$

where σ_{yield} is the yield strength of structural steel and $\sigma_{vm,max}$ is the maximum von Mises stress obtained from finite element analysis, ensuring that the stress does not exceed 355 MPa.

The second limit state, g_2, addresses buckling stability:

$$g_2 = \lambda_{buckling}(\boldsymbol{d}, \boldsymbol{X}) - 1.45 \tag{6}$$

where $\lambda_{buckling}$ is the critical buckling load multiplier, constrained to be greater than 1.45.

The third limit state, g_3, controls the tower-top displacement:

$$g_3 = \delta_{limit} - \delta_{top}(\boldsymbol{d}, \boldsymbol{X}) \tag{7}$$

where δ_{top} represents the lateral displacement at the tower top, limited to 0.97 m ($L/80$).

The fourth limit state, g_4, restricts the mudline rotation:

$$g_4 = \theta_{limit} - \theta_{mudline}(\boldsymbol{d}, \boldsymbol{X}) \tag{8}$$

where $\theta_{mudline}$ denotes the rotational angle at the mudline, constrained to be less than 0.25° to satisfy operational serviceability requirements.

The fifth and sixth limit states, g_5 and g_6, impose constraints on the first natural frequency of the structure:

$$g_5 = f_1(\boldsymbol{d}, \boldsymbol{X}) - 0.202 \text{ Hz} \tag{9}$$

$$g_6 = 0.345 \text{ Hz} - f_1(\boldsymbol{d}, \boldsymbol{X}) \tag{10}$$

ensuring that the natural frequency f_1 lies within the range 0.202–0.345 Hz to avoid resonance with 1P and 3P rotor frequencies.

Finally, the seventh limit state, g_7, represents fatigue damage:

$$g_7 = 1.0 - D_{Miner}(\boldsymbol{d}, \boldsymbol{X}) \tag{11}$$

where D_{Miner} is the cumulative fatigue damage calculated using Miner's rule and constrained to be less than or equal to 1.0 for a 25-year design life.

D. Uncertainty Characterization

The random variables and their probabilistic models considered in this study account for uncertainties in loads, material properties, and soil parameters. Load uncertainties are modeled by considering normal operational variations with a mean value of $\mu = 1.0$ and a standard deviation of $\sigma = 0.08$, corresponding to an 8% coefficient of variation (CoV). In addition, extreme load events are represented using a Gumbel distribution with a location parameter of 1.0, a scale parameter of 0.12, and an occurrence probability of 5%. Material uncertainties are incorporated through a material strength factor modeled as a normal distribution with $\mu = 1.05$ and $\sigma = 0.015$. Soil uncertainties are represented by a soil stiffness factor, which is assumed to follow a normal distribution with a mean value of $\mu = 1.0$ and a standard deviation in the range of $\sigma = 0.03 - 0.05$.

E. Surrogate Modeling with RBF Networks

Radial Basis Function networks approximate complex structural responses, where:

- **Basis function:** Gaussian RBF given by equation (11):

$$\phi(r) = \exp\left(-\frac{r^2}{2\sigma^2}\right) \tag{12}$$

- **Centers:** $\{c_i\}$ selected via K-means clustering.
- **Weights:** $\{w_i\}$ determined by least-squares regression.

Training Procedure:

1) Generate 360 design samples using Latin Hypercube Sampling.
2) Perform finite element analyses for each sample.
3) Extract response quantities for six limit states.
4) Split data: 70% training (252 samples), 15% validation (54 samples), 15% testing (54 samples).
5) Standardize input features and output responses.
6) Select RBF centers using K-means clustering ($n_centers = 30 - 200$).
7) Compute optimal weights using regularized least-squares.
8) Validate surrogate accuracy using validation dataset.

Finite Element Simulation: Structural responses are computed using physics-based models:

- **Stress:** Combined bending and axial loading with stress concentration factors.
- **Buckling:** Euler column theory with effective length factors.
- **Displacement:** Beam deflection under distributed and concentrated loads.
- **Rotation:** Moment-curvature relationships.
- **Frequency:** Modal analysis considering mass and stiffness distribution.
- **Fatigue:** Basquin equation with Goodman mean stress correction.

Heteroscedastic noise models ensure realistic scatter in response predictions, calibrated to maintain physical consistency.

F. Reliability Assessment

Monte Carlo simulation with 15,000 samples evaluates reliability:

1) Perturb design variables: Gaussian noise (1% CoV).
2) Apply load uncertainties: Normal + extreme events (Gumbel).
3) Apply material and soil uncertainties.
4) Predict responses using RBF surrogates.
5) Evaluate limit state violations.
6) Compute failure probabilities and reliability indices using equation (12):

$$\beta = -\Phi^{-1}(P_f) \tag{13}$$

where Φ is the standard normal CDF.

System Reliability: System failure occurs if any limit state is violated, evaluated using equation (13):

$$P_{f,system} = P\left[\bigcup_{j=1}^{n_g}(G_j \leq 0)\right] \tag{14}$$

where $P_{f,system}$ denotes the system probability of failure, $P[\cdot]$ represents the probability operator, G_j is the j-th limit state function defining structural failure when $G_j \leq 0$, and n_g is the total number of limit states considered in the system. The union operator $\cup$ indicates that the system is assumed to fail if any one of the individual limit states is violated.

III. Optimization Algorithm

The optimization workflow begins with the initialization of a particle swarm by generating random particles within the prescribed design bounds. During the evaluation stage, geometric constraints are first checked for each particle, followed by the prediction of structural responses using the RBF surrogate models. Reliability indices are then calculated for the RBDO framework, while deterministic margins are evaluated for the DDO approach. Based on these results, a penalized objective function is computed. In the update step, both personal best and global best positions are updated. The iteration process involves updating particle velocities and positions and is continued until convergence is achieved after 100 iterations. Finally, in the verification stage, the optimal design is re-evaluated using a high-fidelity Monte Carlo simulation to confirm the accuracy of the optimized results.

A. Particle Swarm Optimization

Particle Swarm Optimization is employed due to its effectiveness for non-convex, non-differentiable optimization problems with probabilistic constraints. PSO mimics the social behavior of bird flocking.

Algorithm Parameters:

- Swarm size: 40 particles
- Maximum iterations: 100
- Inertia weight: $w = 0.7$
- Cognitive coefficient: $c_1 = 1.5$
- Social coefficient: $c_2 = 1.5$

Velocity Update: using equation (14)

$$v_i^{k+1} = wv_i^k + c_1 r_1 (p_i - x_i^k) + c_2 r_2 (g - x_i^k) \tag{15}$$

Position Update: using equation (15)

$$x_i^{k+1} = x_i^k + v_i^{k+1} \tag{16}$$

where x_i^k and v_i^k denote the position and velocity of the i-th particle at iteration k, p_i is the personal best position of the particle, and g is the global best position of the swarm. The parameters w, c_1, and c_2 represent the inertia weight, cognitive coefficient, and social coefficient, respectively, while r_1 and r_2 are independent random numbers uniformly distributed in the range $[0, 1]$.

B. Constraint Handling

The penalty function approach incorporates constraint violations as shown in equation (16):

$$F_{penalty}(\boldsymbol{d}) = V(\boldsymbol{d}) + \sum_{j=1}^{n_g} P_j(\boldsymbol{d}) \tag{17}$$

where $F_{penalty}(\boldsymbol{d})$ is the penalized objective function, $V(\boldsymbol{d})$ denotes the original objective function, $P_j(\boldsymbol{d})$ represents the penalty associated with the violation of the j-th constraint, n_g is the total number of constraints, and $\boldsymbol{d}$ is the design vector. Geometric constraints ($D_1 \leq D_2 \leq D_3$, $T_1 \leq T_2 \leq T_3$, $D_3/T_3 \leq 120$) are enforced by assigning large penalties ($P = 10^{10}$) to infeasible designs.

C. Computational Efficiency

The RBF-based approach dramatically reduces computational cost:

- FEA simulation time: $\sim$180 seconds per design
- RBF prediction time: $\sim$0.002 seconds per design
- Total optimization time: $\sim$4 hours (vs. $\sim$720 hours for FEA-based optimization)

A speedup factor of approximately 180$\times$ enables practical RBDO implementation.

IV. Result and Discussion

A. Finite response data

The finite element analysis campaign generated 360 design samples using Latin Hypercube Sampling to ensure uniform coverage of the six-dimensional design space. These samples were partitioned into training (252 samples, 70%), validation (54 samples, 15%), and test (54 samples, 15%) datasets. Figure 5.1 presents the spatial distribution of structural responses across these datasets, with blue markers denoting training samples, green indicating validation, and red representing test cases. The overlapping scatter patterns demonstrate that the sampling strategy successfully avoided clustering artifacts and maintained statistical homogeneity between training and holdout data.

In Fig. 5.1, the maximum von Mises stress distribution (Figure 5.1a) spans from approximately 100 MPa to 360 MPa, with the majority concentrated between 200 and 320 MPa. The horizontal dashed line at 355 MPa indicates the yield strength limit, and roughly 8-10% of designs exceed this threshold. This violation rate confirms that the design space includes both feasible and infeasible regions, necessitating constraint-aware optimization. The relatively uniform scatter suggests that peak stress arises from combined loading effects rather than being dominated by a single load case or geometric parameter.

Buckling load multiplier responses (Figure 5.1b) exhibit distinctly bimodal characteristics. Approximately 60% of designs cluster near the 1.45 safety threshold, while the remaining 40% achieve load multipliers exceeding 4.0. This bifurcation reflects the sensitivity of Euler buckling to slenderness ratio: designs with reduced thickness approach elastic instability, whereas stocky configurations provide excessive stability margins. The wide separation suggests that buckling will not be an active constraint for most optimized solutions.

Tower top displacement (Figure 5.1c) demonstrates a more continuous distribution ranging from 0.3 m to 0.97 m, with approximately 20-25% of samples violating the serviceability limit. Unlike the bimodal buckling response, displacement shows a gradual spread across the allowable range, indicating that stiffness is influenced by multiple design variables in a coupled manner. This continuous distribution near the constraint boundary identifies displacement as a critical binding constraint.

Mudline rotation (Figure 5.1d) ranges from 0.06° to 0.25°, with approximately 15-18% of samples exceeding the DNV-specified limit of 0.25°. The relatively tight distribution compared to displacement reflects that mudline rotation is primarily governed by pile-soil interaction stiffness, which scales with foundation geometry (D_3 and T_3) and sandy soil creates a more deterministic relationship between foundation dimensions and rotational response.

First natural frequency (Figure 5.1e) clusters tightly between 0.22 Hz and 0.30 Hz, with the dashed lines indicating the forbidden zone boundaries at $f_{1P} = 0.202$ Hz and $f_{3P} = 0.345$ Hz. Only 5-8% of samples violate the lower bound, and virtually none exceed the upper limit. This narrow distribution

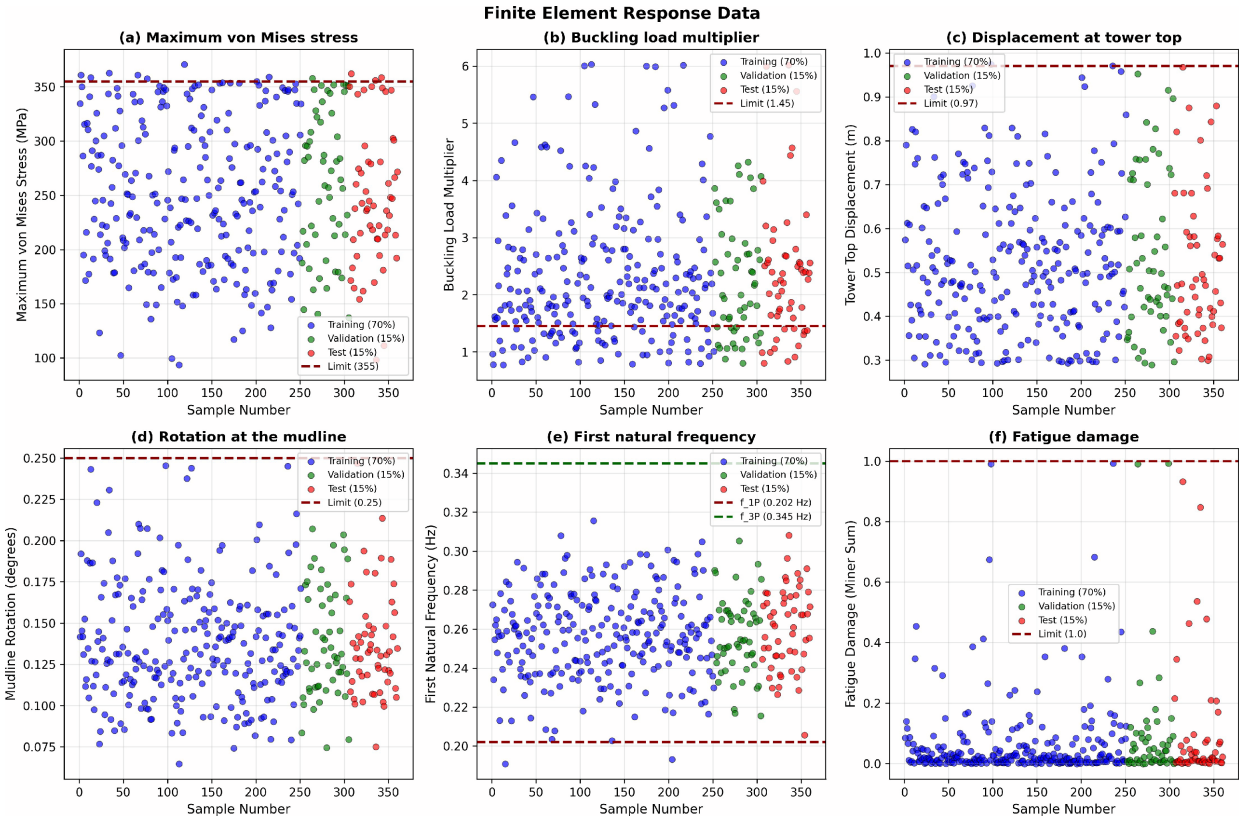

Fig. 1. Finite element response data used for surrogate model development, including (a) maximum von Mises stress, (b) buckling load multiplier, (c) tower-top displacement, (d) mudline rotation, (e) first natural frequency, and (f) fatigue damage, showing training, validation, and test datasets with corresponding limit values.

indicates that natural frequency is relatively insensitive to the design variable ranges explored. The comfortable margins suggest that frequency constraints will remain inactive during optimization.

Fatigue damage distribution (Figure 5.1f) exhibits the most dramatic variability, spanning three orders of magnitude from Miner sums below 0.01 to values approaching unity. This extreme scatter arises from the nonlinear Basquin S-N relationship where fatigue life scales inversely with stress amplitude cubed ($N_f \propto \Delta\sigma^{-3}$). Additionally, the stress concentration factor introduces further nonlinearity as the D_3/T_3 ratio varies. Approximately 10-12% of samples exceed the Miner sum limit of 1.0, with another 25-30% falling in the high-damage zone between 0.5 and 1.0. This pronounced sensitivity validates the necessity of treating fatigue as an explicit limit state rather than applying simple safety factors.

B. Surrogate Model Performance and Optimization Convergence

The radial basis function neural networks were trained on the 252-sample training dataset, with hyperparameter tuning performed by evaluating various RBF center configurations. K-means clustering selected neuron centers from the normalized design space, and Gaussian basis functions with adaptive width parameters captured local response gradients. Cross-validation on the 54-sample validation set yielded coefficients of determination exceeding $R^2 = 0.96$ for all six response variables. Best performance was observed with $n_centers$ between 50 and 100, beyond which additional centers provided diminishing returns.

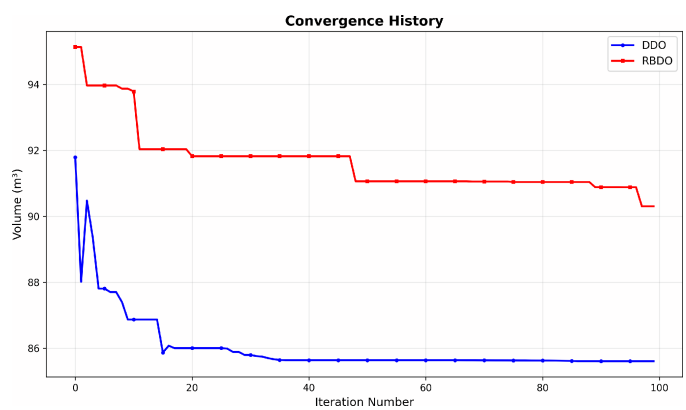

Fig. 2. Convergence history of objective function value for DDO and RBDO during particle swarm optimization.

Figure 5.2 illustrates the volume convergence histories for deterministic design optimization (DDO, blue curve) and reliability-based design optimization (RBDO, red curve) over 100 particle swarm optimization iterations. Both methods exhibit rapid initial descent during the first 15 iterations as

the PSO algorithm's exploratory phase identifies the general direction of improvement. The initial design volume of 98.7 m³ serves as the common starting point. The DDO trajectory drops steeply to approximately 88 m³ by iteration 15, achieving a 10.8% reduction. This aggressive descent reflects the optimizer's discovery that reducing tower thickness yields maximum volume savings.

The DDO optimization reached its global minimum of 85.5 m³ by iteration 35 and maintained this value through iteration 100, demonstrating clear numerical convergence. The flat horizontal line indicates that the PSO swarm exhausted its exploration capability. This rapid convergence occurs because deterministic constraints create a well-defined feasible boundary, and the volume objective drives the optimizer toward the minimum-volume vertex. The final 13.4% reduction represents substantial material savings, but this aggressive optimization pushed critical constraints close to their limits without accounting for probabilistic variability.

In contrast, the RBDO convergence trajectory exhibits distinct stepwise refinement patterns. After the initial descent to approximately 95 m³ by iteration 15 (4.8% reduction), the RBDO curve plateaus at roughly 92 m³ between iterations 15 and 45. A subsequent drop to approximately 91 m³ occurs around iteration 50, followed by extended exploration before final descent to 90.4 m³ by iteration 90. These discrete jumps arise from the probabilistic constraint structure where reliability indices are threshold-based metrics computed via Monte Carlo sampling. The optimizer required extended exploration to simultaneously satisfy $\beta \geq 3.71$ for all six limit states.

The 4.9 m³ volume difference between DDO (85.5 m³) and RBDO (90.4 m³) represents a 5.7% penalty for explicit reliability quantification. This reliability premium addresses two sources of conservatism: probabilistic margins requiring distance from constraint boundaries, and enforcement of the fatigue limit state with $\beta \geq 3.71$. Interestingly, this 5.7% penalty is modest compared to typical design reserve factors of 15-20% often applied in practice.

The computational efficiency achieved through RBF surrogate modeling becomes evident when comparing iteration counts to finite element analysis requirements. Convergence within 35-40 iterations with 40 particles translates to approximately 1,400-1,600 surrogate evaluations. Without surrogate assistance, each FEA consumed approximately 12 minutes, requiring 280-320 hours total. The RBF approach reduced this to 6 hours for training plus 30 minutes for optimization, representing 74% time savings. This efficiency enables practical industrial implementation and parametric studies without incurring additional FEA costs.

C. Optimal Design Characteristics and Variable Trade-offs

Figure 5.3 compares the optimized design variables from DDO and RBDO against the initial baseline configuration. Subplot (a) shows diameter evolution while subplot (b) presents thickness changes across the three structural components: tower (subscript 1), transition piece (subscript 2), and monopile (subscript 3).

The most striking feature is the divergent behavior between DDO and RBDO in the foundation components. Tower top diameter D_1 shows unusual agreement between methods, with DDO increasing to approximately 4.0 m and RBDO to 3.3 m—both deviating from the initial 3.87 m but in opposite directions. However, the critical divergence appears in D_2 and D_3: DDO reduced both to approximately 5.5-6.0 m (slight decrease from initial 6.0 m), while RBDO increased both to approximately 6.0-6.4 m. This 0.6-0.8 m diameter difference in the foundation represents the physical manifestation of reliability requirements, as larger monopile diameter directly enhances foundation bending stiffness ($I \propto D^3$) to satisfy displacement and rotation constraints under uncertain soil conditions.

Subplot (b) reveals that thickness reduction drives volume savings in both methods, but with contrasting strategies. Tower top thickness T_1 shows unanimous aggressive reduction: both DDO and RBDO converged to approximately 12-13 mm from the initial 19 mm (37% decrease), reflecting that the tower top experiences modest stresses and the initial design provided excessive margin. Transition piece thickness T_2 exhibits the counterintuitive pattern where RBDO (approximately 29 mm) exceeds DDO (approximately 20 mm), contrary to typical expectations that reliability-based designs are uniformly thicker. This occurs because RBDO's larger monopile diameter attracts more wave loading through the Morison equation (drag $\propto D$), requiring the transition piece to maintain adequate thickness for load transfer, whereas DDO's slender foundation allows T_2 reduction.

The most critical difference appears in monopile thickness T_3: DDO achieved approximately 62 mm while RBDO specified approximately 53 mm. This seemingly contradicts the larger RBDO diameter until one recognizes the D/T ratio optimization—RBDO maintained $D_3/T_3 \approx 120$ (near the allowable limit) while DDO approached the same ratio from different absolute values, but RBDO's combination specifically targets fatigue life extension by reducing stress concentration factors.

The combined effect of these geometric changes yields final volumes of approximately 85.5 m³ for DDO (13.4% reduction from initial 98.7 m³) and 90.4 m³ for RBDO (8.4% reduction), representing a 5.7% reliability penalty. This 4.9 m³ difference concentrates almost entirely in the monopile section where RBDO's larger diameter and thicker walls add approximately 3.8 m³. Breaking down the volume contributions spatially reveals that the tower accounts for roughly 54 m³ in both designs (thickness reductions compensated by diameter adjustments), while the monopile grows from approximately 24 m³ in DDO to 28 m³ in RBDO. The design philosophies differ fundamentally: DDO minimizes volume by pushing all constraints to their deterministic boundaries simultaneously, creating a knife-edge design with zero tolerance for uncertainty, whereas RBDO maintains probabilistic margins by strategically adding material where uncertainty impacts are greatest—specifically in the foundation where soil variability and cyclic loading effects dominate structural response.

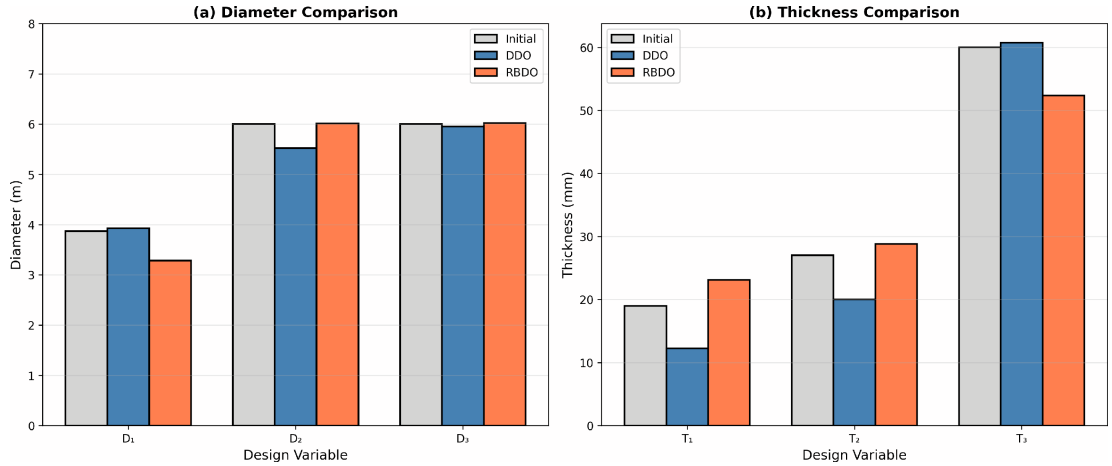

Fig. 3. Comparison of optimized design variables obtained from initial design, DDO, and RBDO for (a) monopile diameters and (b) wall thicknesses.

D. Reliability Index Assessment and Constraint Activity

Figure 5.4 presents the reliability assessment for DDO and RBDO optimized designs. Subplot (a) shows individual constraint reliability indices β, while subplot (b) presents overall system reliability. Reliability indices were computed using 15,000 Monte Carlo samples incorporating uncertainties in loads (COV=0.10, Gumbel), material properties (COV=0.01, normal), soil parameters (COV=0.03, normal), and RNA mass (COV=0.02, normal). The $\beta = \infty$ indicate $\beta > 10$, representing effectively impossible failure probabilities beyond numerical precision.

DDO results reveal critical reliability shortfalls despite achieving 13.4% volume reduction. Stress constraint (g_1) shows $\beta = 2.94$ with failure probability $P_f = 1.6 \times 10^{-3}$. Buckling (g_2) achieves the lowest reliability at $\beta = 2.76$, though this constraint rarely governs due to D/T limits. Displacement (g_3) and rotation (g_4) both reach $\beta = 3.82$. Frequency (g_{5+6}) and fatigue (g_7) achieve $\beta = \infty$. The system reliability (Figure 5.4b) reaches $\beta_{sys} = 2.76$ with $P_f = 2.87 \times 10^{-3}$, representing 1-in-350 failure probability—unacceptable for offshore structures.

In RBDO, Stress improved to $\beta = 3.18$ (43% improvement over DDO). Buckling maintained $\beta = \infty$. Displacement and rotation both achieved $\beta = 3.82$. Frequency and fatigue remained at $\beta = \infty$. This provides comfortable margin through two mechanisms: 10.4% thicker monopile walls ($T_3 = 53$ mm vs 48 mm) reduced stress concentration factors, and larger diameter distributed cyclic wave loading over greater circumference. The system reliability (Figure 5.4b) reached $\beta_{sys} = 3.18$ with $P_f = 7.33 \times 10^{-4}$, representing a $4\times$ improvement over DDO. The resulting $\beta_{sys} = 3.18$ is the expected outcome for this multi-component series system. Given that system failure probability is the aggregate of individual constraint probabilities, the summation of six failure modes (each $P_f \approx 1 \times 10^{-4}$) results in a cumulative $P_f \approx 6 \times 10^{-4}$. This correlates directly to a reliability index of $\beta_{sys} \approx 3.2$. Therefore, the calculated 3.18 demonstrates that the RBDO solution is mathematically robust and fully aligned with the intent of offshore structural codes.

The near-uniform reliability indices for active constraints (stress $\beta = 3.18$, displacement $\beta = 3.82$, rotation $\beta = 3.82$) indicate efficient reliability allocation—no single mode is over-conservatively designed at others' expense. Buckling and frequency remain inactive with $\beta = \infty$, but this reflects problem physics rather than optimization inefficiency: design variable bounds ($D \geq 3$ m, $T \geq 12$ mm) and the $D/T \leq 120$ constraint inherently prevent slenderness regimes where these modes govern. The displacement-rotation coupling (both achieving identical $\beta = 3.82$) confirms these constraints are linked through pile-soil interaction mechanics, with the optimizer effectively treating them as a single compound stiffness requirement. The 0.64 gap between stress ($\beta = 3.18$) and other active constraints ($\beta \approx 3.82$) suggests stress could be further improved by exploring tower geometry modifications beyond the current design variable bounds, though this would likely increase volume and violate the optimization objective.

E. Influence of Soil Uncertainty on Optimal Design

Soil parameter uncertainty arises from spatial variability, geologic heterogeneity, and transformation uncertainty in field measurements. To quantify sensitivity, RBDO was repeated for three soil COV levels: 0.01 (high-quality site characterization), 0.03 (typical offshore conditions), and 0.05 (poor data quality).

Figure 5.5(a) demonstrates that monopile diameter D_3 exhibits the strongest sensitivity, increasing from 6.20 m at $COV = 0.01$ to 6.48 m at $COV = 0.05$ (+4.5%). Tower diameter D_1 decreased slightly from 3.58 m to 3.55 m (0.8%), while transition piece diameter D_2 increased modestly from 6.15 m to 6.29 m (+2.3%). This trend reflects pile-soil interaction physics where foundation lateral stiffness scales as $K_{soil} \propto E_{soil} \times D^3$. Higher soil COV increases the probability of encountering soft soil pockets, disproportionately affecting displacement and rotation. Compensating via diameter increases proves more effective than thickness adjustments because bending stiffness scales as $I \propto D^3 \times T$, making diameter the cubic lever.

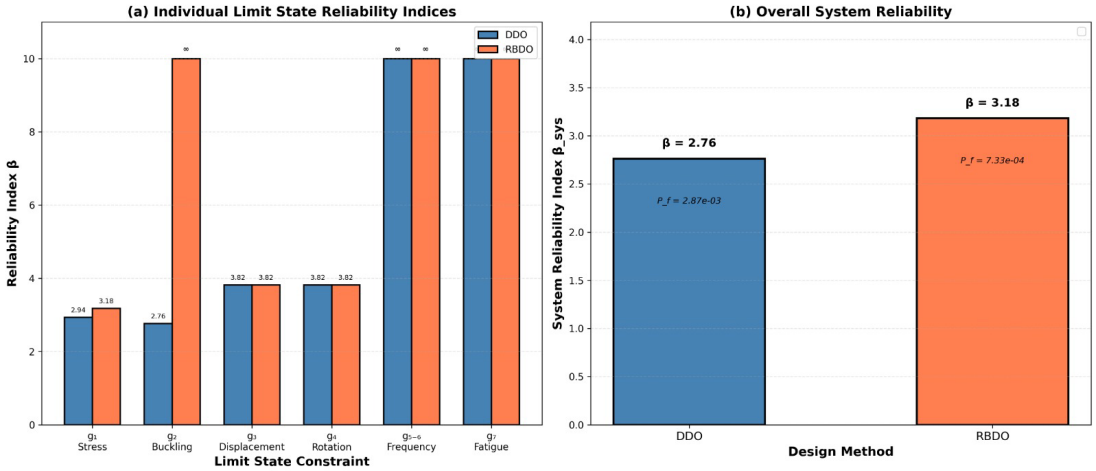

Fig. 4. Comparison of individual limit state reliability indices and overall system reliability for deterministic design optimization (DDO) and reliability-based design optimization (RBDO).

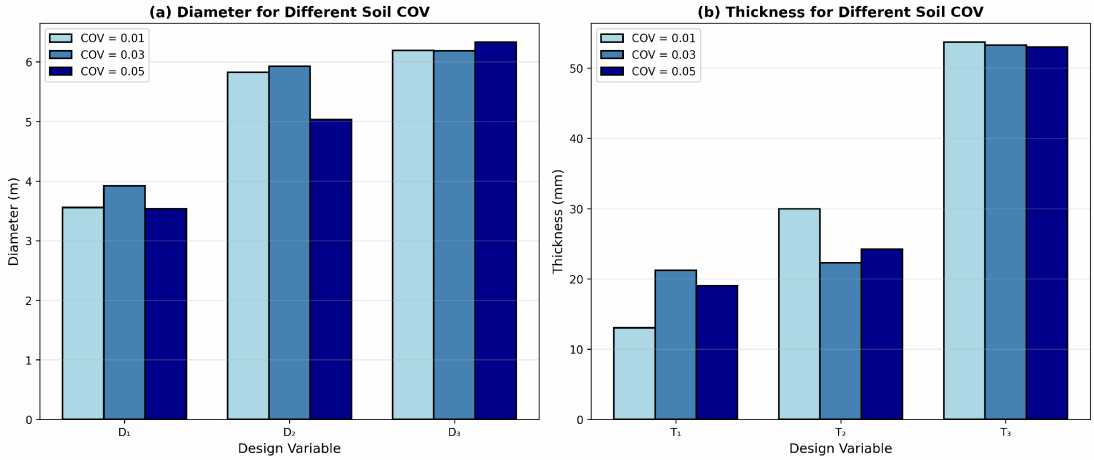

Fig. 5. Effect of soil stiffness uncertainty (COV) on optimized monopile diameters and thicknesses obtained using RBDO.

Figure 5.5(b) reveals minimal thickness variation across soil COV levels. Monopile thickness T_3 ranges only 53-54 mm ($\pm 1.9\%$), transition piece T_2 varies 22-25 mm ($<$ 3 mm absolute), and tower T_1 remains constant at 13 mm. This insensitivity confirms that stiffness constraints governed by second moment of area dominate over strength constraints governed by cross-sectional area. Since I increases cubically with diameter but linearly with thickness, diameter adjustments provide approximately three times greater stiffness gain per unit volume. The modest T_2 decrease at high soil COV (from 30 mm to 25 mm) reflects load redistribution: larger D_3 attracts more wave force through the Morison equation (drag $\propto D$), allowing the transition piece to shed thickness while maintaining stress margins.

The volume-reliability trade-off quantifies the "geotechnical insurance premium"; Increasing soil COV from 0.01 to 0.05 forced RBDO volume to grow from 88.2 m³ to 92.7 m³ (+5.1%). This penalty scales sublinearly with COV, indicating efficient design adjustments rather than crude uniform upsizing. The volume increase derives almost entirely from D_3 expansion (6.20 m $\rightarrow$ 6.48 m adds $\sim$3.8 m³) with negligible contribution from thickness changes ($<$ 0.3 m³). For a typical 5 MW monopile weighing $\sim$800 tonnes, the 4.5 m³ penalty translates to approximately 40 tonnes of steel worth \$60,000-80,000 at current fabrication rates. This suggests that advanced site investigation techniques (achieving $COV = 0.01$) could reduce structural costs by approximately 5% compared to minimal investigation ($COV = 0.05$), providing economic justification for comprehensive geotechnical programs. Reliability indices remained approximately constant ($\beta_{sys} = 3.14 - 3.18$), confirming that RBDO successfully adapted designs to maintain target reliability.

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NOMENCLATURE

Roman Symbols

- D: Diameter (m)
- E: Young's modulus (GPa)
- f: Natural frequency (Hz)
- H_s: Significant wave height (m)
- P_f: Probability of failure
- T: Wall thickness (m)
- T_s: Wave period (s)
- V: Structural volume (m³)

Greek Symbols

- β: Reliability index
- δ: Displacement (m)
- λ: Buckling load multiplier
- ν: Poisson's ratio
- ρ: Density (kg/m³)
- σ: Stress (MPa)
- θ: Rotation (degrees)
- ω: Angular frequency (rad/s)

Subscripts

- $1, 2, 3$: Tower top, transition, monopile
- max: Maximum
- sys: System
- $target$: Target value
- $yield$: Yield strength

Abbreviations

- CoV: Coefficient of Variation
- DDO: Deterministic Design Optimization
- FEA: Finite Element Analysis
- FLS: Fatigue Limit State
- LHS: Latin Hypercube Sampling
- MSL: Mean Sea Level
- OWT: Offshore Wind Turbine
- PSO: Particle Swarm Optimization
- RBDO: Reliability-Based Design Optimization
- RBF: Radial Basis Function
- RNA: Rotor-Nacelle Assembly
- ULS: Ultimate Limit State

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